DECISION BRIEF · OPEN-WEIGHT DOCUMENT AI ON YOUR OWN HARDWARE

Which open-weight model should read your documents, and what will it cost to run?

Research proposals, procurement packets, contracts and reports keep their most useful facts in tables, charts, scans and clauses. I tested eleven open-weight models on public stand-ins for that work, confirmed the leaders on data they had never seen, and priced the hardware to run them privately.

THE SHORT ANSWER

Use small specialised models, not one big one. A 0.9-billion-parameter table reader beat a 32-billion-parameter vision model on financial tables. The recommended document-reading models all fit on one 24 GB data-centre GPU, an L4-class server planned at about $6,000 to $10,000. Keep a person reviewing amounts and contract answers: no model recovered 99% of the numbers.

See the recommendations | **Download this brief (PDF)** | Explore every comparison →

Tayler Erbe · Data scientist, applied AI. I design model evaluations that end in a decision: which model, on what hardware, at what cost. About this work

01 / FINDINGS

What to use for each job

Each recommendation names the job, the model, why it was chosen, the evidence and the main limit. Every card links to the comparison behind it.

BUDGET AND FINANCIAL TABLES

PaddleOCR-VL-1.6

Confirmed · pilot first

Rebuilt financial tables more faithfully than a 32B vision model, and kept the numbers.

Evidence Table similarity 0.852 against 0.762, and 97.4% of reference numbers recovered against 93.2%, on 100 unseen financial-report tables. It passed a test written down before scoring.

Limit 97.4% is below a 99% bar, so amounts still need review.

Next Check it on your own budget tables.

See the table comparison →

WHOLE-PAGE TEXT

Nemotron 3.5 Parse

Lead · check numbers

The most accurate page text of the models tested, and the fastest.

Evidence Word overlap 0.899 against 0.820 for the 32B model, the same share of numbers (88.2% against 88.3%), and 140 items a minute against 9.

Limit On 80 pages, the planned numbers test could not rule out a small loss. That makes it a lead, not a confirmed win.

Next Pair it with a numbers check on pilot documents.

See the page comparison →

CHARTS

Qwen3-VL, with the method chosen per model

Method matters

The same instruction helped one model and cut another's accuracy by about 43 percentage points.

Evidence On 1,322 unseen chart questions: Qwen3-VL 32B with three-answer voting 84.6%. The 8B model 75.0% at baseline, but 32.3% when asked to re-check itself.

Limit Public charts. Questions about the same chart are related.

Next On 24 GB or less, use the 8B at baseline. With 48 GB, use the 32B with voting.

See the chart comparison →

CONTRACT CLAUSES

Qwen3 32B; gpt-oss-20b for smaller GPUs

Human review required

Best on contract questions that ask whether a document supports a statement.

Evidence ContractNLI: 18 of 30 correct, against 14 for gpt-oss-20b and 12 for a hosted model. On CUAD, none beat the 15 of 30 scored by always answering "absent".

Limit Modest scores on 30 questions per task. Not legal advice.

Next Decide whether the gap justifies the larger GPU. See Hardware.

See the contract comparison →

CROPPED TEXT REGIONS

Qwen3-VL 8B

Small is enough

Read cropped paragraphs as accurately as the 32B models.

Evidence 1.2% character error on 50 cropped paragraphs, on a 12 GB laptop GPU.

Limit Exploratory set of 50 crops.

See the text-region comparison →

FINDING TABLES AND FIGURES

DocLayout-YOLO

Keep the detector

A dedicated layout detector found figures far more reliably than the page parser.

Evidence Figure-finding score 0.876 against 0.612 on 120 pages.

Limit For tables, the two were not clearly different.

See the layout comparison →

Why the top score is not always the recommendation

In some comparisons another setup has a higher text score. The PaddleOCR official pipeline, for example, had the best word overlap on pages (0.922), but it recovered only 77.1% of the numbers. These recommendations favour keeping numbers, because budgets, awards and invoices fail on numbers first.

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02 / HARDWARE AND BUDGET

Start from the budget

Each tier shows what it can run, whether that was measured or estimated, and a dated price range for a complete system, not just the graphics card. Prices were checked in October 2026 and move quickly. Refresh them before buying.

EXISTING COMPUTER, NO GPU

$0 extra

Uses what you have

- Read text and spreadsheets directly when the original file exists
- Basic OCR and the review screens

Not benchmarked here. A starting point, not a scanned-document solution.

16 GB GPU WORKSTATION

$1,500 to $2,500

One small model at a time

- Either parser (about 1 to 2 GB of weights)
- Qwen3-VL 8B (about 10 GB)

Fit estimated from weight sizes; not tested. Planning allowance from the hardware guide.

12 GB MOBILE WORKSTATION · TESTED

About $4,650 new

RTX 4000 Ada Laptop GPU, 64 GB RAM

- Measured: Qwen3-VL 8B and Qwen2.5-VL 7B on tables, charts, text regions and pages, one model at a time

Retail list price for a comparable configuration, October 2026. Refurbished units cost far less.

32 GB DESKTOP, RTX 5090 · TESTED

About $3,600 to $5,300

Where every final-round model was measured

- Measured: all parsers and vision models, the 32B models, gpt-oss-20b
- The two recommended parsers handled 73 to 140 items a minute; the 32B vision model 9

Complete prebuilt systems listed in 2026. Graphics-card shortages move these prices.

SINGLE L4 SERVER, 24 GB · TARGET

$6,000 to $10,000

Data-centre card, low power

- Fits: both parsers, Qwen3-VL 8B, gpt-oss-20b for contracts
- Tight: any 32B model

Fit from weight sizes; not benchmarked. Dated planning allowance, not a quote.

SINGLE L40S SERVER, 48 GB · TARGET

$16,000 to $21,000

Room for a large model beside a parser

- Fits: everything recommended, including Qwen3 32B for contracts
- Headroom for several requests at once

Fit from weight sizes; not benchmarked. Dated planning allowance, not a quote.

L4 or L40S?

The L4 covers the document-reading routes: tables, pages, charts, text regions and layout. The L40S mainly buys the 32B contract model and headroom. On the public contract questions the 32B led the 20B on ContractNLI (18 against 14 of 30) and they tied on CUAD. Whether that gap is worth about $10,000 more depends on your contract fields and the volume you need to process. A short pilot can measure both before you buy.

Speeds were measured on the 32 GB desktop. They compare models fairly but are not speeds for an L4 or L40S. Purchase price leaves out engineering, review time, maintenance and energy. Running models on your own hardware keeps documents in your environment, but it does not make the system secure by itself. Details: hardware and budget guide.

03 / RESULTS

The clearest results

Financial tables: small reader vs 32B

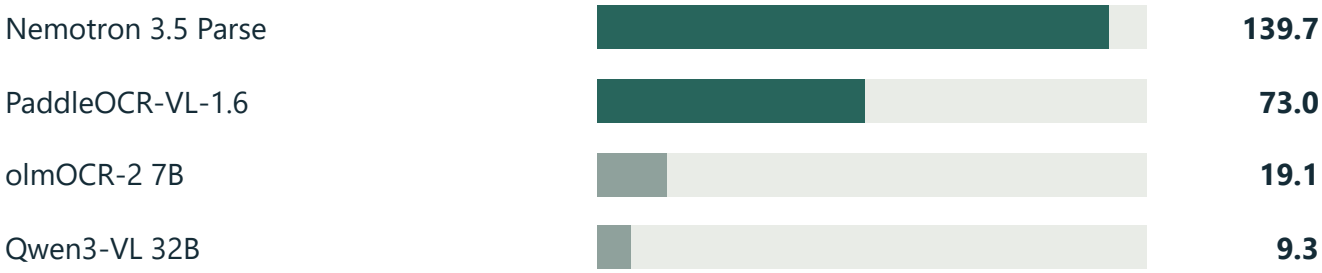
100 unseen financial-report tables. Same output limit for both. Higher is better.



Passed the pre-registered test. Table similarity measures structure, not fully correct tables.

Speed: items processed per minute

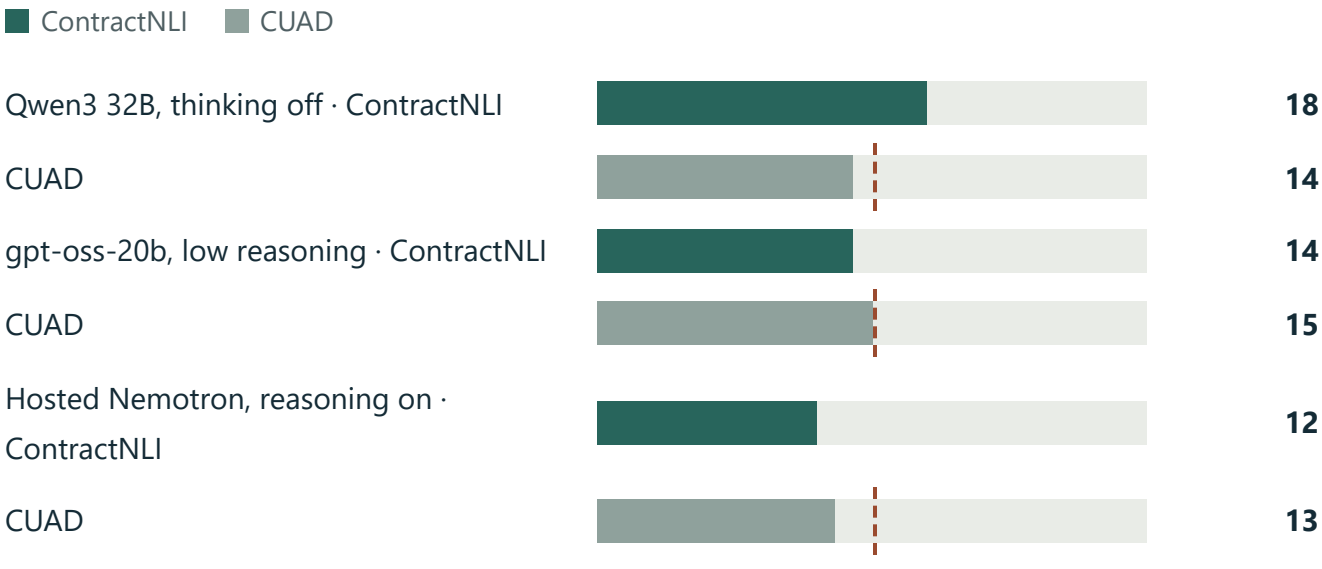
One batch of 20 pages and 20 tables, four requests at a time, same desktop.



Relative speeds on one machine, not a promise for other hardware.

Contract questions: correct out of 30

The same 60 public documents for every configuration. The dashed line is the CUAD score from always answering "absent" (15 of 30).



CUAD does not separate these models. Two more configurations are in the full contract comparison.

Explore all 18 comparisons

04 / METHOD

How the decision was made, and how I would make it for you

The method matters more than any single score. It is the same six steps whether the documents are research awards, procurement packets or contracts.

1

Define the jobs

Split the documents into the kinds of reading they need, and name the failure that matters for each. Here, that was losing numbers.

2

Freeze fair tests

Pick public stand-ins with answer keys, fix the questions and the scoring code, and let no model grade another.

3

Shortlist by budget

Choose candidates that fit the hardware you can afford, from small parsers to 32B models.

4

Explore, then confirm

Explore on development data, write down what will count as a win, then test once on data no model has seen.

5

Price the fit

Map the winning setup to a hardware tier and a dated budget, separating measured speed from estimates.

6

Validate on real documents

Label a representative set of your own documents and measure field accuracy before approving anything.

Why it took several rounds

Each round answered a question the previous one raised. None of them simply reran tests until a good number appeared.

Round 1

Vision models

QUESTION

Which local vision model reads each part of a document, on a 12 GB laptop and a 32 GB desktop?

TEST

Five vision models and several prompting methods on frozen public sets for text, tables, charts and pages.

FINDING

The 8B model matched the 32B on text and tables. Prompting methods helped some models and hurt others.

DECISION

Route by model and task. Test the promising methods properly.

Round 2

Pre-registered

QUESTION

Do the chosen methods hold on data no model has seen?

TEST

Three-answer voting on 1,322 new chart questions and layout masking on 80 new pages, with the pass rules written first.

FINDING

Voting helped the 7B and 32B, not the 8B. Masking cleaned up text but lost numbers.

DECISION

Keep voting optional, keep masking experimental, and look for specialist tools.

Contract track

Text models

QUESTION

Can models answer contract questions with evidence a reviewer can check?

TEST

CUAD and ContractNLI public contracts, quoting versus selecting evidence by ID, and a controlled reassessment of a hosted model.

FINDING

Selecting evidence by ID removed copy failures. The hosted model's poor first score was a configuration problem.

DECISION

Compare every candidate on the same 60 documents.

Round 3

New candidates

QUESTION

Do specialist document parsers or gpt-oss-20b beat the incumbents?

TEST

Three parsers on every earlier page and table set, and gpt-oss-20b on the contract sets.

FINDING

Leads emerged: PaddleOCR-VL for tables, Nemotron Parse for pages. They were found by looking, so they were not yet proof.

DECISION

Write the two hypotheses down and test them on fresh data.

Round 3b and 3c

Confirmation

QUESTION

Do the leads hold on unseen data, with the same output limit for every model?

TEST

100 new financial tables and 80 new pages, the parsers' official pipelines, budget checks, and Qwen3 32B on the 60 contract documents. 3c reran two arms that had failed for software reasons.

FINDING

The table lead was confirmed. The page lead held on text but was inconclusive on numbers. Official pipelines did not beat a single pass.

DECISION

Close the evaluation, route by task, and validate on real documents.

Why it stopped: the leads were either confirmed or bounded. More public benchmarks would add little, so the next evidence has to come from real documents. Full methods and definitions →

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05 / NEXT STEPS

From evaluation to a working service

A mixed PDF is not one task. The proposed service sends each part of a document to the tool that did best on that kind of reading, keeps every answer linked to its source page, and leaves the decision with a reviewer.

Receive

Keep the original file untouched.

Identify

Find pages, tables, charts and text, and use native text where it exists.

Route

Send each part to its recommended model.

Combine

Assemble the results into one record, linked to source pages.

Review

Show proposed values beside the page. Flag numbers and unread parts.

Export

Send approved data to search, reports or other systems.

Clearing a backlog vs keeping up

A first pass over an existing archive can run in batches: load one model, process every waiting table, then switch, so a single GPU is never asked to hold everything at once. After that, new and changed documents arrive in smaller volumes. How long the backlog takes cannot be estimated from these tests. It depends on pages per file, how often each route is needed, the target server and review effort, and the pilot should measure it.

A bounded pilot

1. Agree the fields that matter and where the documents come from.
2. Label about 100 representative documents, including difficult scans, with two reviewers on a subset.
3. Run the recommended routes and measure field accuracy, numbers, traceability, time and correction effort.
4. Decide whether to scale, adapt or fine-tune, using the measured errors.

Fine-tuning a model such as Qwen3-VL 8B on your own labelled documents may help. It was not tested here, and it cannot fix text that was never read in the first place.

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06 / REPORTS AND DATA

Read it, share it, check it

Reports

Final advisory · PDF
Recommendations, confirmed tests, limits.

Hardware and budget guide · PDF
Tiers, dated prices, what was not measured.

Implementation roadmap · PDF

The next phase after the evaluation.

Hosted model reassessment · PDF
Why a poor first score was a configuration problem.

Routing report, rounds 1 and 2 · PDF
Historical, 21 October 2026.

Data

Every comparison on this site (JSON)
The explorer's data, built from the files below.

Final aggregates (JSON)
With source commit and checksums.

Paired statistics (JSON)
Intervals and the pre-registered tests.

All result rows (JSON)
Every scored configuration, with sources.

Data audit
How missing values were restored and checked.

Code

Repository
Test harness, frozen protocols, scorers and tests.

Evidence and methods page
The detailed record behind this brief.

ABOUT THIS WORK

Tayler Erbe

I am a data scientist working in applied AI. I design evaluations that turn model results into decisions: which model to use for each job, what hardware it needs, what it costs, and what still has to be proven. This study used public benchmarks only. Local models did all of the document reading and answering. AI coding assistants helped build the test harness, and no model graded another.

To talk about an evaluation for your own documents, see the case study on my site or get in touch.

Last updated 29 October 2026.

Public benchmarks only. No private documents, labels or model responses appear on this site.

Source and frozen protocols